

Change overview changes
after November 5th meeting and additional remarks

INTRODUCTION

This document contains a summary of the changes, as presented during the Ereg Topic Group XII meeting November 5th, to the Draft version 2.0 Initial Vehicle Information that was previously published on the Ereg site. With these changes, the 2.0 version becomes final.

```
<xs:annotation>
  <xs:documentation>Message:   Initial Vehicle Information</xs:documentation>
  <xs:documentation>Version:   2.0</xs:documentation>
  <xs:documentation>Generated: 2025-12-18 12:00</xs:documentation>
</xs:annotation>
```

OVERVIEW OF CHANGES IVI 2.0 VERSION

CONSOLIDATEDMAXIMUMNETPOWERELECTRICENGINE

De tag has been renamed to make it more consistent with ConsolidatedMaximum30MinutesPower

```
<xs:element name="ConsolidatedMaximumNetPowerElectric" minOccurs="0"
type="Decimal62"/>
```

NUMBEROFDOORSREAR

De tag NumberOfDoorsRear has been renamed to NumberOfDoorsBack.

AXLEGROUPMEMBERTABLE

Corrected : not mandatory anymore.

TYRETYPE INFORMATION

In the previous IVI version 2.0 the TyreType information contains combinations of TyreTypes as separate codes.

For instance XLATMS being a combination of:

XL = Extra Load – The tire is reinforced to carry more weight than a standard tire of the same size.

AT = All Terrain

M+S = Mud and Snow – Indicates basic suitability for muddy and snowy conditions

A new table is added to be able to combine different TyreTypes without having to add enumerations for different combinations to the list.

```
<xs:complexType name="TyreTypeTable">
  <xs:sequence>
    <xs:element name="TyreTypeGroup" maxOccurs="unbounded" type="TyreTypeGroup"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="TyreTypeGroup">
  <xs:all>
    <xs:element name="TyreType" minOccurs="0" type="TyreTypeCode"/>
  </xs:all>
</xs:complexType>
```

With the following enumerations mentioned:

```
<xs:simpleType name="TyreTypeCode">
  <xs:restriction base="xs:string">
    <xs:maxLength value="7"/>
    <xs:enumeration value="AT"/>
    <xs:enumeration value="AL"/>
    <xs:enumeration value="AS"/>
    <xs:enumeration value="HLC"/>
    <xs:enumeration value="MC"/>
    <xs:enumeration value="MS"/>
    <xs:enumeration value="RF"/>
    <xs:enumeration value="S"/>
    <xs:enumeration value="XL"/>
    <xs:enumeration value="MST"/>
    <xs:enumeration value="3PMSF"/>
    <xs:enumeration value="IGS"/>
    <xs:enumeration value="PAX"/>
  </xs:restriction>
</xs:simpleType>
```

Also new code has been added:

HLC = High load capacity

ENUMERATIONS

LocationMarkingsVehiclePartSideCode (fieldlength to: 10)

From	To
FRONT	FRONTSIDE
LEFT	LEFTSIDE
RIGHT	RIGHTSIDE
BACK	BACKSIDE

LocationMarkingsVehiclePartSideSectionCode (fieldlength to: 10)

From	To
MIDLE	MIDDLE

EuCountryCode Norway, Switzerland, United Kingdom and Iceland added:

CH	Switzerland
NO	Norway
IS	Iceland
UK	United Kingdom
LI	Liechtenstein

ERRORS IN TAGNAMES

Name:	Changed to:
ElongatedCabineIndicator	ElongatedCabIndicator
AerisolIndicator	AerosolIndicator

ProtectionPenetratingObjectsForestryIndicator	ProtectionPenetratingObjectsForestryIndicator
ProtectionPenetratingObjectsNonForestryIndicator	ProtectionPenetratingObjectsNonForestryIndicator
FallingObjectProtectiveStructureForestryIndicator	FallingObjectsProtectiveStructureForestryIndicator
FallingObjectProtectiveStructureNonForestryIndicator	FallingObjectProtectiveStructureNonForestryIndicator
AerodynamicDeviceEquipped	AerodynamicDeviceEquipped
AerodynamicDeviceEquippedCode	AerodynamicDeviceEquippedCode
MechanicalCouplingNumberOfTrackTrainCouplingPoint	MechanicalCouplingNumberOfTrackTrainCouplingPoint
CabineCategoryCode	CabCategoryCode
SoundLevelDriverPerceivedOpenCabine	SoundLevelDriverPerceivedOpenCab
SoundLevelDriverPerceivedClosedCabine	SoundLevelDriverPerceivedClosedCab
WhscHc	WhscThc
WhtcHc	WhtcThc

RIMOFFSET

Fielddefinition corrected.

```
<xs:simpleType name="RimOffset">
  <xs:restriction base="xs:decimal">
    <xs:totalDigits value="6"/>
    <xs:fractionDigits value="3"/>
    <xs:minInclusive value="-999.999"/>
    <xs:maxInclusive value="999.999"/>
  </xs:restriction>
</xs:simpleType>
```

IVIREFERENCEID

Additional check on the IVIReferenceId to make sure that a GUID UUID version 4 is being used. The min/maxlength has been removed, as this is covered by the new pattern check:

```
<xs:simpleType name="IviReferenceId">
  <xs:restriction base="xs:string">
    <xs:pattern value="[0-9a-fA-F]{8}-[0-9a-fA-F]{4}-4[0-9a-fA-F]{3}-[89abAB][0-9a-fA-F]{3}-[0-9a-fA-F]{12}"/>
  </xs:restriction>
</xs:simpleType>
```

VECTOGROUP

While fuel type (e.g., diesel, LNG, electricity) influences the engine map and fuel properties used in the simulation, VECTO does not calculate CO₂ emissions per fuel type in isolation.

Also the O category vehicles (trailers) are included in the VECTO (Vehicle Energy Consumption Calculation Tool) framework under EU regulations.

This is why it has been decided to move the Co2 Vecto fields from EnergySource to VectoGroup:

SpecificCo2EmissionsMass
SpecificCo2EmissionsPerson

TECHNICALLYPERMISSIBLEMAXIMUMSTATICVERTICALMASSCOUPLINGPOINT

The location of the vertical maximum static mass has been moved from the mechanical coupling table to its own table. As a result, the maximum static vertical mass can be defined as follows:

1. At vehicle level, when no coupling is installed, but towing maximum masses are available. For example, a passenger car that is delivered without a coupling device.
2. At coupling level, by means of the MechanicalCouplingNumber as the identifier, which is also included in the mechanical coupling table. This allows the maximum static vertical mass to be linked to multiple attachment points on the vehicle. For example, when a vehicle has two coupling device options, such as a 50 mm ball coupling (75 kg) and a drawbar coupling (1,000 kg).

The S-value of the coupling device itself remains in the MechanicalCouplingTable.

PERMISSIBLE HEIGHT

For agricultural vehicles a range for height of incomplete vehicles has been added:

MaximumPermissibleHeightMin
MaximumPermissibleHeightMax

Scope

This Regulation applies to motor vehicles of categories M₁, M₂, M₃, N₁, N₂ and N₃, as well as trailers of categories O₃ and O₄ as specified in Article 4 of Regulation (EU) 2018/858, including those designed and constructed in one or more stages, and to systems, components and separate technical units intended for such vehicles, and C₁, C₂ and C₃ class tyres, as specified in UN Regulation No 117 ⁽²³⁾, with the exception of ice grip tyres.

47. EXHAUSTEMISSIONLEVEL

The following enumerations have been added:

Euro 7 FL
Euro 7 FE
Euro 7 GL
Euro 7 GE
Euro 7 GS
Euro 7 GT
Euro 7 HL
Euro 7 HE
Euro 7 HS
Euro 7 HT
Euro 7 TL
Euro 7 TE
Euro 7 TS
Euro 7 TT

48.3 OBM TESTFAMILY

The OBM testfamily will be registered in TestFamilyIdentifiers table. Other testfamilies will be added as well for Euro 7 in future. This could also be the case for batteries and brakes (to be determined). The paper format of the CoC also contains the OBM family identifier in entry 48.3. and might contain more identifiers that are not an approval number.

Extend enumeration for TestFamilyIdentifierCode with: OM

48.3 IMPLEMENTING ACTS

A new table has been added for registration of the implementing acts related to Euro 7. For instance, the main implementing act (MIA) can be added here, with a possibility to add other subject approvals as well.

```
<xs:complexType name="SubjectApprovalTable">
  <xs:sequence>
    <xs:element name="SubjectApprovalGroup" maxOccurs="unbounded" type="SubjectApprovalGroup"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="SubjectApprovalGroup">
  <xs:all>
    <xs:element name="SubjectApprovalCertificateNumber" type="Text80"/>
  </xs:all>
</xs:complexType>
```

MECHANICALCOUPLING

MECHANICALCOUPLINGPARTSTABLE

A mechanical coupling consists of several parts, whereby the different parts of the coupling can also have their own data. This applies to the vehicle categories: M, N, O, T, C, R and S.



As requested the option to add information on parts of a mechanical coupling has been added to the IVI structure:

```
<xs:complexType name="MechanicalCouplingPartsTable">
  <xs:sequence>
    <xs:element name="MechanicalCouplingPartsGroup" maxOccurs="unbounded" type="MechanicalCouplingPartsGroup"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="MechanicalCouplingPartsGroup">
  <xs:all>
    <xs:element name="MechanicalCouplingPartsTypeAccordingAppendixEU" minOccurs="0" type="Text120"/>
    <xs:element name="MechanicalCouplingPartsMake" minOccurs="0" type="Text52"/>
    <xs:element name="MechanicalCouplingPartsManufacturersTypeDesignation" minOccurs="0" type="Text40"/>
    <xs:element name="MechanicalCouplingPartsApprovalMark" minOccurs="0" type="Text80"/>
    <xs:element name="MechanicalCouplingPartsApprovalNumber" minOccurs="0" type="Text50"/>
    <xs:element name="MechanicalCouplingPartsValueD" minOccurs="0" type="Decimal62"/>
    <xs:element name="MechanicalCouplingPartsValueDc" minOccurs="0" type="Decimal62"/>
    <xs:element name="MechanicalCouplingPartsValueV" minOccurs="0" type="Decimal62"/>
    <xs:element name="MechanicalCouplingPartsValueS" minOccurs="0" type="Integer6"/>
    <xs:element name="MechanicalCouplingPartsValueU" minOccurs="0" type="Decimal62"/>
  </xs:all>
</xs:complexType>
```

MECHANICALCOUPLINGCANBEFITTEDTABLE

To make it possible to specify multiple couplings that can possibly be attached to a vehicle for incomplete vehicles with their own ldata, a table has been added and the fields “MechanicalCouplingCanBeFittedType” en “MechanicalCouplingCanBeFittedClass” have been moved to “MechanicalCouplingCanBeFittedTable”.

```
<xs:complexType name="MechanicalCouplingCanBeFittedGroup">
  <xs:all>
    <xs:element name="MechanicalCouplingCanBeFittedType" minOccurs="0" type="Text80"/>
    <xs:element name="MechanicalCouplingCanBeFittedClass" minOccurs="0" type="MechanicalCouplingCanBeFittedClassCode"/>
    <xs:element name="MechanicalCouplingCanBeFittedValueD" minOccurs="0" type="Decimal62"/>
    <xs:element name="MechanicalCouplingCanBeFittedValueDc" minOccurs="0" type="Decimal62"/>
    <xs:element name="MechanicalCouplingCanBeFittedValueV" minOccurs="0" type="Decimal62"/>
    <xs:element name="MechanicalCouplingCanBeFittedValueS" minOccurs="0" type="Integer6"/>
    <xs:element name="MechanicalCouplingCanBeFittedValueU" minOccurs="0" type="Decimal62"/>
  </xs:all>
</xs:complexType>
```

SIGNING (MINIMUM) REQUIREMENTS - CONCLUSION

Manufacturers, in minimum, have to apply an **advanced electronic seal** to the IVI messages which can be made with a **non-qualified certificate**. Article 36 of regulation 910/2014 states the requirements for such a seal.

Manufacturers **can** (but are not required to) apply a **qualified electronic seal** which can be made with a **qualified certificate**.

This seal must be applied as of 5 July 2026

Having a valid code signing certificate issues by **trusted CA** and applying xmldsig will suffice to meet these requirements for submission of IVI messages.

X-ADES can be applied, as best practice; however this is not mandatory.